

# Homework #2 (AI Agent)

Area Baroud

## Lebanon Electronics Component Finder Agent

### Part C: Reporting

#### 1. The Research

An agent is an AI system that can independently execute tasks by making decisions and taking actions on its own to complete a task. It chooses when and how to use different tools, and take actions to achieve a goal by itself

Types: • **ReAct (Reason + Act)**: Reason → Act → Observe  
agent analyses the problem and decides what to do      ↓      uses a tool or performs an action      → checks result and continues

- **Reflexion** (adds self-critique after acting)  
completes a task, evaluates what went wrong, creates feedback for itself, uses that feedback to improve
- **Plan and Execute**  
agent plans all steps upfront, then executes each step

Tools: A function the model can call, with a name/description/inputs. They are external sources to perform tasks that it cannot do by itself. Without them, there is no reliable math, or the ability to take actions in the real world, it can only generate responses based on its learned knowledge.

Loop: input → Reason → act → observe → Repeat  
agent receives goal from user      ↓      thinks about the task and decides what needs to be done      ↓      takes an action (ex: use a tool)      ↓      receives result of that action  
if goal not achieved, it reasons again, take another action and observes the result



The model is only the reasoning step, it doesn't run the loop, the code does. Memory is not built into it, it is added as a separate component (short-term or long-term).

## 2. The idea behind your choice

As a CCE student in Lebanon, I find it hard to find specific electronics components, there are no organized online inventories. I usually have to search multiple supplier websites, check availability and verify specifications manually. I found it interesting to use AI to make this process smarter by allowing the agent to search, compare and recommend components instead of manually doing everything myself.

## 3. Your design decisions

I deliberately turned off the library's automatic function calling feature so I could understand how AI agents actually work internally. I wanted to see the reasoning process behind each action. Also, I used DuckDuckGo search over a paid search API because it provides a free way to access web search without requiring an API subscription, given how limited my time was. In addition, I used a maximum of 6 loop iterations as a safety limit, so that the program won't keep running forever if the agent got stuck.

## 4. How your agent works

Tool 1: search\_component\_stores

↳ This tool takes a component name, searches for it on the web with "Lebanon", so without it the agent will have no information about real stores, and wouldn't include Lebanese stores

Tool 2: check\_distance

↳ takes 2 locations: store area and my location, so it's a separate search from the first tool. It answers how far is it from me. Without it, you'd have no idea which one is actually close and worth going to.

Tool 3: save\_summary

↳ takes everything found by the first 2 tools and writes it into a .txt file. Because everything said inside that chat disappears once the conversation ends, I added this tool so in case I want to actually buy the component found, the store name will still be reachable.

The loop: ① My code looks at what I'm asking and decides to



either request tools or provide a final response

- ② If it requests a tool, it executes that specific function and sends the outcome back to Gemini, then prompts it again
- ③ This keeps repeating until it has sufficient information and responds with a straight-forward answer.

So, nobody tells it in advance do exactly these 3 steps in this exact order, it figures it out.

## 5. Reflection

### What was hard?

Getting the model I wanted was hard, it would work for a short period of time then it randomly stops working and gives an error saying "Model not available" or "Daily free requests have exceeded". So I had to keep switching to different model names to get it running again

### What surprised me?

I added a line that printed out exactly what my tools found before Gemini formed it into a sentence. But looking at the raw results, all it had were generic link (Google maps homepage) with no actual value, it gave a specific value despite having no way of knowing the exact distance. This is a real issue called Hallucination.

### What I'd improve with more time

I'd connect the distance tool to an actual maps tool for exact numbers since if I don't tell the agent where I am, it guesses a location instead of asking me directly.